

An Auditable Complexity-Aware PAC-Bayesian Framework for Linear and Takagi–Sugeno Time-Series Forecasting under Temporal Dependence

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Abstract—PAC-Bayesian certificates for time-series predictors can be mathematically valid yet operationally misleading when the prior reuses certification data, the loss is left unbounded, or model-selection choices are not charged. This paper presents an auditable computational framework for linear and first-order Takagi–Sugeno (TSK) forecasting with overlapping, potentially nonstationary windows. The framework specializes an established time-uniform supermartingale PAC-Bayes result; it does not claim a new general concentration theorem. A chronological segment constructs every candidate prior, and an explicit hierarchical code charges model family, lag, ridge penalty, optional radius, realized rule count, localization variant, and prior scale. The guarantee directly covers clipped Gibbs risk and, by convexity, the clipped Bayesian model average; deterministic posterior-center forecasts are evaluated empirically and use the certificate only as a screening signal.

A 30-series synthetic development study separates structural rule reduction from sparse activation. Relative to a fixed 12-rule TSK model, radius-controlled TSK reduced randomized dimension, Gaussian KL, and certificate value, with paired confidence intervals excluding zero. A sensitivity analysis over $K \in \{2, 3, 4, 6, 8, 12\}$ shows that the large contrast with $K = 12$ is mainly a rule-count effect: small fixed rule bases can match or tighten the radius-controlled certificate, and radius control does not significantly dominate the best fairly charged fixed K . Actual multi-rule examples from SETAR and Tetouan are reported to show what the learned structures represent. A permissive Tetouan deployment gate failed, whereas a stricter gate frozen before an independent four-region PJM evaluation improved RMSE and asymmetric cost over a seven-day seasonal-naïve fallback. Secondary post-hoc SARIMA and ETS benchmarks, designed after the PJM outcomes were opened, often match or outperform the deployed models and therefore further limit the operational claim. The evidence supports an auditable integration of structural complexity, risk screening, validation margins, and abstention; it does not establish universal superiority of radius selection or nonlinear fuzzy forecasting.

Index Terms—PAC-Bayesian learning, Takagi–Sugeno fuzzy systems, time-series forecasting, martingales, temporal dependence, structural sparsity, risk certificates.

I. INTRODUCTION

Time-series predictors are commonly selected by validation error and assessed on a held-out horizon. This workflow measures empirical forecasting performance, but it does not by itself quantify the conditional risk of a data-dependent model when lagged examples overlap, the observations are temporally dependent, and the underlying process may change over time. The limitation is especially relevant for flexible forecasting

systems whose complexity depends on several jointly selected structural and numerical choices.

Takagi–Sugeno (TSK) systems represent nonlinear dynamics through convex combinations of local affine rules. Their behavior depends on the lag order, antecedent geometry, number of rules, and regularization of the consequent parameters. These choices affect accuracy, interpretability, and effective statistical complexity. In many forecasting studies, however, the rule count and consequent dimension are treated as implementation details rather than explicit components of a generalization certificate.

PAC-Bayesian analysis provides a natural language for relating empirical loss to a prior–posterior complexity term. In sequential forecasting, a certificate can nevertheless be misleading if the prior is localized with certification observations, squared loss is treated as bounded without justification, or lag, regularization, radius, and rule-count choices are selected without paying their complexity cost. Moreover, a non-vacuous certificate need not identify the predictor with the lowest raw test RMSE or the best operational cost.

This study develops an auditable computational framework for linear and first-order TSK forecasting under overlapping and potentially nonstationary windows. The time-uniform supermartingale argument used in the analysis is established in the PAC-Bayes literature and is not presented as a new concentration theorem. The methodological contribution lies in the chronological construction of the candidate priors, explicit hierarchical charging of the complete model index, analytic evaluation of a Gaussian empirical-risk upper bound, separation of structural sparsity from activation sparsity, and integration of the certificate into a frozen model-selection and abstention protocol.

The central claim is deliberately limited. We do not introduce a new fuzzy clustering algorithm, prove that radius-based rule construction dominates all fixed rule bases, or show that TSK models universally outperform linear and seasonal predictors. Instead, we ask whether complete removal of rule blocks makes TSK complexity auditable and whether a valid certificate can serve as one conservative screening component inside a broader deployment rule.

The contributions are as follows.

- We define a chronological candidate family with complete structural index $M = (F, p, \gamma, r, K)$, where F denotes the model family, p the lag, γ the ridge penalty, r an optional radius, and K the realized rule count. Scaling,

antecedent geometry, prior means, and prior variances are constructed only from the initial segment.

- We provide a computable bounded-loss specialization of the established time-uniform supermartingale PAC-Bayes framework [1]. The contribution is the measurable linear/TSK specialization, augmented hierarchical prior, analytic empirical-risk upper bound, and audited implementation.
- We distinguish structural rule-count reduction from sparse activation. A synthetic development study compares ridge, fixed- K TSK, radius-controlled all-active TSK, and radius-controlled Top-3 TSK. A subsequent sensitivity analysis over $K \in \{2, 3, 4, 6, 8, 12\}$ separates the effect of reducing rule count from any additional benefit of radius construction.
- We expose actual posterior-center fuzzy rules from a multi-rule SETAR model and from the Tetouan development model. These examples support inspection of the learned local structures without treating them as physical truth or as directly certified deterministic predictors.
- We report a negative Tetouan development gate and an internally locked PJM holdout evaluation. Secondary SARIMA and exponential-smoothing benchmarks are explicitly labeled post hoc because they were designed after the PJM outcomes were opened and do not modify the frozen gate or its original claim.

The evidence supports an integration claim rather than a fuzzy-model superiority claim. Structural rule reduction substantially lowers dimension, KL, and certificate value relative to an unnecessarily dense 12-rule TSK model. Small fixed rule bases can, however, match or tighten the radius-controlled certificate, and ridge remains the easiest family to certify on average. The independent energy decisions mostly select linear predictors, while the secondary statistical baselines show that stronger forecasting alternatives can improve further on the deployed models. The certificate therefore functions as an auditable risk-screening quantity, not as a direct guarantee of raw RMSE, future stability, or model-class dominance.

The remainder of the paper reviews the relevant literature, defines the chronological model family and the three prediction objects used in the study, formalizes the augmented hierarchical prior, describes the development and operational protocols, and reports the structural, interpretability, and decision results.

II. RELATED WORK AND POSITIONING

A. PAC-Bayes under Dependence and Sequential Observation

PAC-Bayes for dependent observations is established. Chromatic and blocking arguments cover dependency graphs and stationary mixing processes [2]; oracle inequalities have been developed for weakly dependent time-series prediction [3], [4]; and recent results address Markov models and stable nonlinear dynamical systems [5], [6]. These approaches typically require explicit dependence, stationarity, boundedness, or stability assumptions.

Martingale and supermartingale methods provide a different route. Seldin *et al.* derived PAC-Bayesian inequalities for martingales [7]. Online and time-uniform constructions

subsequently allowed data-dependent posterior sequences while keeping the prior predictable [1], [8]. We use this route because overlapping lag windows form an adapted sequence, and because it avoids treating an estimated mixing coefficient as known. The bounded-loss formula used here is a direct specialization of this literature, not a new general concentration inequality.

B. Computable and Non-Vacuous Certificates

A formal inequality becomes useful only when the prior, posterior, and empirical term can be computed without hidden data reuse. Data-dependent priors can be valid when their information source is explicitly separated or otherwise controlled [9], [10]. Work on neural-network certificates shows that prior localization and posterior-scale selection are decisive for tightness [11]. Foong *et al.* show that small-sample PAC-Bayes bounds may remain looser than direct test bounds even after careful optimization [12]. Heavy-tail and unbounded-loss PAC-Bayes results offer alternatives to clipping [13]–[15]; the current implementation does not claim those guarantees and instead reports clipping explicitly.

C. Sparse TSK Systems

Bayesian TSK models, block-sparse dynamic fuzzy models, and sparse rule or coefficient learning are well developed [16]–[21]. Clustering-based rule construction and online TSK identification are also classical [22], [23]. Existing work generally targets prediction, parameter estimation, or interpretability; it does not provide the specific audited chain from a prior-only radius cover to a realized randomized dimension, KL cost, and sequential certificate.

D. Precise Novelty Claim

The present study does not introduce a new clustering algorithm, a new general PAC-Bayes theorem, or a proof that TSK models dominate linear or seasonal forecasts. Its contribution is the intersection of: (i) a chronologically valid finite prior over complete forecasting structures; (ii) a computable sequential certificate for a linear-in-consequents TSK model; (iii) a controlled fixed- K versus radius-controlled structural ablation; and (iv) an operational protocol that records a negative development case before an independent frozen confirmatory case. Table I makes the distinction from the closest PAC-Bayes literature explicit.

III. SEQUENTIAL SETUP AND MODEL FAMILY

A. Chronological Roles

Let $(\mathcal{F}_t)_{t \geq 0}$ be the data filtration and

$$X_t = (Y_{t-1}, \dots, Y_{t-p})^\top \quad (1)$$

be $\mathcal{F}_{t-1}$ -measurable. Every trajectory is divided, without shuffling, into $\mathcal{D}_{\text{prior}}$, $\mathcal{D}_{\text{bound}}$, $\mathcal{D}_{\text{validation}}$, and $\mathcal{D}_{\text{test}}$. The first segment constructs all candidate-specific priors. The bound segment fits posterior centers, the validation segment selects among the predeclared candidate/posterior family, and the test segment is opened only after selection or deployment

TABLE I

POSITIONING RELATIVE TO REPRESENTATIVE PAC-BAYES LITERATURE. “THIS WORK” USES A KNOWN TIME-UNIFORM SUPERMARTINGALE ROUTE; ITS CONTRIBUTION IS THE CHRONOLOGICAL, COMPUTABLE SPECIALIZATION AND THE STRUCTURAL/OPERATIONAL EVALUATION.

Work	Data regime	Loss / guarantee	Main computational focus	Difference from the present study
Seldin et al. [7]	Martingales	PAC-Bayes martingale inequalities	General concentration	Provides foundational martingale PAC-Bayes inequalities; no chronological TSK construction or rule-count audit.
Chugg et al. [1]	Sequential, non-IID	Time-uniform PAC-Bayes recipes	General supermartingale framework	The bound used here is a bounded-loss specialization, not a competing general theorem.
Pérez-Ortiz et al. [11]	IID classification	Tight neural-network risk certificates	Joint prior/posterior optimization	Motivates computable non-vacuous certificates, but not temporal dependence or fuzzy structure.
Foong et al. [12]	IID, small samples	Tightness limits	Bound-vs-test comparison	Motivates reporting vacuity and clipping; it does not address sequential dependence.
Haddouche-Guedj [14]	Sequential/heavy-tailed	Supermartingale bounds for heavy tails	Unbounded-loss control	Offers a route beyond clipping; the current implementation deliberately uses bounded clipped loss.
Alquier–Wintenberger [3] and Eringis et al. [6]	Weakly dependent time series / stable systems	Generalization under dependence	Rates or stability/mixing conditions	These require dependence or stability assumptions; the primary certificate here uses predictability and bounded loss instead.
This work	Overlapping, possibly nonstationary forecast windows	Conditional sequential clipped risk	Hierarchical structural coding; radius-vs-fixed- K ablation; frozen deployment gate	Empirically separates structural sparsity from activation sparsity and tests a negative development gate followed by an independent confirmatory energy case.

decisions are frozen. Certification uses bound plus validation observations; test observations are excluded.

All target scaling parameters are computed from $\mathcal{D}_{\text{prior}}$. This makes the standardized lag vector, antecedent geometry, and clipping interval measurable at the start of the certification horizon.

B. Ridge and TSK Predictors

The ridge family uses the design $\phi_p(x) = (1, x^\top)^\top$. A first-order TSK system with K rules uses Gaussian memberships

$$\mu_{kj}(x_j) = \exp\left[-\frac{(x_j - c_{kj})^2}{2s_{kj}^2}\right], \quad (2)$$

$$\alpha_k(x) = \frac{\prod_j \mu_{kj}(x_j)}{\sum_{\ell=1}^K \prod_j \mu_{\ell j}(x_j)}, \quad (3)$$

and prediction

$$f_{M,a}(x) = \sum_{k=1}^K \alpha_k(x)(a_{k0} + a_k^\top x). \quad (4)$$

With fixed antecedents, the predictor is linear in the stacked consequent vector a . Its design vector is

$$\phi_M(x) = [\alpha_1(x)(1, x^\top), \dots, \alpha_K(x)(1, x^\top)]^\top, \quad (5)$$

with randomized dimension $D_M = K(p+1)$. Ridge is the $D_M = p+1$ linear family and is kept separate in the structural prior.

C. Certified and Operational Prediction Objects

The study distinguishes three prediction objects that should not be conflated. First, the *Gibbs predictor* samples $a \sim Q_M$ and incurs the clipped loss in (12); Proposition IV.2 directly controls its conditional sequential risk. Second, the *clipped Bayesian model average* is

$$f_Q^{\text{BMA}}(x) = \mathbb{E}_{a \sim Q_M} [\Pi_{[-B,B]} f_{M,a}(x)], \quad (6)$$

and is covered by Corollary IV.3 through convexity. Third, the deterministic posterior-center point forecast used for validation, test RMSE, and the deployment studies is

$$f_M^{\text{point}}(x) = \Pi_{[-B,B]} f_{M,\mu_{Q,M}}(x). \quad (7)$$

Before projection, linearity in a gives $\mathbb{E}_Q[f_{M,a}(x)] = f_{M,\mu_{Q,M}}(x)$. After projection, however, $\mathbb{E}_Q[\Pi f_{M,a}(x)]$ need not equal $\Pi f_{M,\mu_{Q,M}}(x)$. Consequently, the certificate is not presented as a direct guarantee for the deterministic point forecast. In the operational studies it is one eligibility signal combined with validation margins and temporal-stability checks; the reported raw RMSE and cost remain empirical holdout quantities.

D. Prior-Only Antecedent Constructions

For radius-controlled TSK, standardized prior inputs are covered by deterministic farthest-first centers until all prior points lie within radius r or the predeclared rule cap is reached. Candidates that hit the cap before coverage are ineligible. Rule spreads are estimated from prior assignments with fixed numerical floors and ceilings. The resulting $K = K(r, \mathcal{D}_{\text{prior}})$ is therefore random before conditioning, but is $\mathcal{G}_0$ -measurable after the prior segment.

The fixed- K baseline uses deterministic farthest-first construction with exactly $K_{\max}$ centers. Radius-controlled all-active TSK retains all rule weights. Radius Top-3 TSK uses the same antecedents but keeps only the three largest firing strengths per sample before renormalization. This latter operation changes computation and occasionally prediction, but not D_M ; it is therefore an activation-sparsity ablation rather than structural sparsity.

E. Complete Model Index and Ridge Fits

The discrete model index is

$$M = (F, p, \gamma, r, K), \quad (8)$$

where F identifies ridge, fixed- K TSK, radius TSK, or the development-only Top-3 family. The radius is absent for ridge and fixed- K candidates. For every predeclared M , the prior and posterior centers use the same ridge penalty γ :

$$\mu_{P,M} = (\Phi_{P,M}^\top \Phi_{P,M} + \gamma I)^{-1} \Phi_{P,M}^\top y_P, \quad (9)$$

$$\mu_{Q,M} = (\Phi_{B,M}^\top \Phi_{B,M} + \gamma I)^{-1} \Phi_{B,M}^\top y_B. \quad (10)$$

Thus γ is not selected silently after the prior is fixed: all candidate-specific prior centers are functions only of the configuration and $\mathcal{D}_{\text{prior}}$. Computing them lazily in software is mathematically equivalent to constructing the whole finite family at $\mathcal{G}_0$.

The validation ordering is RMSE, then MAE, consequent dimension, rule count, radius, ridge penalty, and lag. A separate development analysis minimizes the certificate instead of validation RMSE; this is not the primary predictive selection rule.

F. Bounded Certified Loss

Let

$$B = c_B \max \left\{ 1, \max_{t \in \mathcal{D}_{\text{prior}}} |Y_t| \right\}, \quad (11)$$

where $c_B \geq 1$ is fixed before certification. Define $\bar{Y}_t = \Pi_{[-B,B]} Y_t$ and $\bar{f}_h(X_t) = \Pi_{[-B,B]} f_h(X_t)$. The certified loss is

$$\ell_t(h) = \frac{(\bar{Y}_t - \bar{f}_h(X_t))^2}{4B^2} \in [0, 1]. \quad (12)$$

Target and forecast clipping rates are reported separately. The operational energy studies use $c_B = 1.25$; the synthetic development study uses $c_B = 1$.

IV. COMPUTABLE SEQUENTIAL PAC-BAYES CERTIFICATE

A. Augmented Hierarchical Prior

Condition on the information $\mathcal{G}_0$ available after $\mathcal{D}_{\text{prior}}$. The hypothesis space is augmented with the discrete labels used to construct the Gaussian component:

$$h = (M, v, s, a), \quad M = (F, p, \gamma, r, K), \quad (13)$$

where v denotes the prior-localization variant, s the prior-scale component, and a the consequent vector. The prediction depends on (M, a) ; the labels (v, s) make the mixture accounting explicit.

Let π_F , π_p , and π_γ be uniform on the predeclared family, lag, and ridge-penalty grids. For ridge and fixed- K families, the radius factor is one; for radius-controlled families, $\pi_{r|F}$ is uniform on the radius grid. TSK structures additionally receive the normalized geometric rule-count mass

$$\pi_K(K) = \frac{\exp(-\eta K)}{\sum_{j=1}^{K_{\max}} \exp(-\eta j)}, \quad \eta = \log 2. \quad (14)$$

The structural mass used by the implementation is therefore

$$\pi(M) = \pi_F(F) \pi_p(p) \pi_\gamma(\gamma) \pi_{r|F}(r | F) \pi_{K|F}(K | F), \quad (15)$$

where $\pi_{K|F} = 1$ for ridge and $\pi_{K|F} = \pi_K$ for TSK families. For a radius candidate, $K = K(r, \mathcal{D}_{\text{prior}})$ is fixed after conditioning on $\mathcal{G}_0$. Ineligible radius candidates receive zero executable mass. For each radius-controlled family,

$$\sum_r \pi_{r|F}(r | F) \pi_K(K(r)) \mathbf{1}\{r \text{ eligible}\} \leq 1, \quad (16)$$

and for any evaluated fixed- K subset, $\sum_{K \in \mathcal{K}_{\text{exec}}} \pi_K(K) \leq 1$. The unused family mass is assigned to a dummy hypothesis. Thus the prior is proper without renormalizing after observing which candidates are executable; the reported code lengths are conservative.

For localization variant v and prior scale s , let $\omega(v)$ and $\xi(s)$ denote the predeclared mixture masses. The operational studies use a single chronological localized variant, while the development localization ablation uses two equally weighted variants. The scale set is $\mathcal{S} = \{0.5, 1, 2, 4, 8\}$ with uniform mass $\xi(s) = 1/5$. For every (M, v, s) , define

$$P_{M,v,s} = \mathcal{N}(\mu_{P,M,v}, \text{diag}(s^2 b_{M,v}^2)). \quad (17)$$

The full prior on the augmented space is

$$P(dM, dv, ds, da) = \pi(M) \omega(v) \xi(s) P_{M,v,s}(da), \quad (18)$$

with the remaining structural mass placed on the dummy component.

After observing the certification data, a selected posterior has the form

$$Q = \delta_{\hat{M}} \delta_{\hat{v}} \delta_{\hat{s}} Q_{\hat{M}, \hat{v}, \hat{s}, \hat{\rho}}, \quad (19)$$

where

$$Q_{M,v,s,\rho} = \mathcal{N}(\mu_{Q,M}, \text{diag}(s^2 \rho^2 b_{M,v}^2)), \quad (20)$$

and $\rho \in \{0.01, 0.03, 0.1, 0.3, 1\}$. Because the component labels are explicit coordinates of the augmented hypothesis, the divergence decomposes exactly as

$$\begin{aligned} \text{KL}(Q \| P) = & -\log \pi(\hat{M}) - \log \omega(\hat{v}) - \log \xi(\hat{s}) \\ & + \text{KL}\left(Q_{\hat{M}, \hat{v}, \hat{s}, \hat{\rho}} \parallel P_{\hat{M}, \hat{v}, \hat{s}}\right). \end{aligned} \quad (21)$$

The final term is the exact diagonal-Gaussian divergence

$$\frac{1}{2} \sum_{j=1}^{D_M} \left[\log \frac{\sigma_{P,j}^2}{\sigma_{Q,j}^2} - 1 + \frac{\sigma_{Q,j}^2 + (\mu_{Q,j} - \mu_{P,j})^2}{\sigma_{P,j}^2} \right]. \quad (22)$$

Selection of ρ needs no additional discrete union penalty because Proposition IV.2 is simultaneous over all $Q \ll P$. Temperature selection is different: it is a choice among concentration inequalities and is charged through the temperature mass $\nu(\lambda)$. The supplementary material gives the proper-mass completion and the componentwise derivation in full.

B. Time-Uniform Bounded-Loss Specialization

For $h = (M, a)$, write $\mu_t(h) = \mathbb{E}[\ell_t(h) \mid \mathcal{G}_{t-1}]$ and

$$\hat{R}_t(Q) = \frac{1}{t} \sum_{i=1}^t \mathbb{E}_Q[\ell_i], \quad (23)$$

$$R_t^{\text{seq}}(Q) = \frac{1}{t} \sum_{i=1}^t \mathbb{E}_Q[\mu_i]. \quad (24)$$

Assumption IV.1 (Predictable bounded loss). *For each fixed h , the forecast at time t is $\mathcal{G}_{t-1}$ -measurable and $0 \leq \ell_t(h) \leq 1$ almost surely.*

Proposition IV.2 (Bounded-loss time-uniform specialization). *Let P be $\mathcal{G}_0$ -measurable and let Λ be a finite positive temperature grid with mass function ν . Conditionally on $\mathcal{G}_0$, with probability at least $1 - \delta$, simultaneously for every $t \geq 1$, every $\mathcal{G}_t$ -measurable $Q_t \ll P$, and every $\lambda \in \Lambda$,*

$$R_t^{\text{seq}}(Q_t) \leq \hat{R}_t(Q_t) + \frac{\lambda}{8} + \frac{\text{KL}(Q_t \parallel P) + \log[1/(\delta\nu(\lambda))]}{\lambda t}. \quad (25)$$

Proposition IV.2 is the conditional-Hoeffding instance of the established PAC-Bayes supermartingale recipe [1], [7]. For fixed h , $\exp\{\lambda \sum_{i \leq t} (\mu_i - \ell_i) - t\lambda^2/8\}$ is a nonnegative supermartingale. Mixing over P , applying Ville's inequality, then the Donsker–Varadhan change of measure proves the statement; a union bound charges the finite temperature grid. The proof and the prior-mixture completion are given in the supplementary material.

The reported value is the minimum untruncated right-hand side over the fixed temperature grid, clipped at one only for presentation. Familywise studies allocate δ/S to each of S predeclared series.

C. Analytic Empirical-Risk Upper Bound

For fixed antecedents, $f_{M,a}(x) = \phi_M(x)^\top a$ before projection. Therefore

$$\mathbb{E}_Q[(\bar{Y} - \phi^\top a)^2] = (\bar{Y} - \phi^\top \mu_Q)^2 + \phi^\top \text{diag}(\sigma_Q^2) \phi. \quad (26)$$

Projection cannot increase squared distance to $\bar{Y} \in [-B, B]$. Using the concavity of $u \mapsto \min(1, u)$ gives the pointwise upper bound

$$\hat{r}_i(Q) = \min \left\{ 1, \frac{(\bar{Y}_i - \phi_i^\top \mu_Q)^2 + \phi_i^\top \text{diag}(\sigma_Q^2) \phi_i}{4B^2} \right\}. \quad (27)$$

The implementation substitutes $t^{-1} \sum_i \hat{r}_i(Q)$ for $\hat{R}_t(Q)$, which is conservative.

Corollary IV.3 (Clipped Bayesian model average). *The clipped Bayesian model-average risk is no larger than the Gibbs risk in Proposition IV.2, by convexity of squared loss.*

D. Scope of the Guarantee

The guarantee is for average conditional risk of the clipped Gibbs predictor, and by convexity its clipped model average, over the observed certification horizon. It is valid for adapted overlapping windows and does not require an estimated mixing

coefficient. It is not a guarantee for raw RMSE, an arbitrary deterministic parameter-mean implementation, the future test horizon, or performance relative to a seasonal baseline. Those quantities remain empirical evaluations and are deliberately separated from the certificate.

V. EXPERIMENTAL PROTOCOL

A. Synthetic Development and Structural Ablation

The development study uses six processes and five seeds per process, for 30 complete trajectories. Each trajectory retains 2,000 observations after a burn-in of 500 and is split chronologically 20/45/15/20 into prior, bound, validation, and test roles. Table II gives the data-generating definitions; complete initialization and parameter details are provided in the supplementary material and configuration files.

The original structural ablation compares ridge, fixed- K all-active TSK with $K = 12$, radius-controlled all-active TSK, and radius-controlled Top-3 TSK. Candidate lags are $\{3, 5, 10, 15, 20\}$, ridge penalties are $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, and radii are $\{0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 4, 5\}$. The rule cap is 12. The certificate uses temperatures $\{0.125, 0.25, 0.5, 1, 2, 4\}$, the prior scales and posterior ratios in Section IV, and familywise confidence $\delta = 0.05/30$.

Candidate rows contain no test metric. For each family and selection strategy, the selected candidate is serialized before the synthetic test role is opened. The original four-family engine evaluated 12,700 eligible candidates with no build failures.

B. Exploratory Fixed- K Sensitivity and Statistical Inference

To avoid interpreting the $K = 12$ contrast as evidence for radius construction itself, an explicitly exploratory extension evaluates fixed $K \in \{2, 3, 4, 6, 8, 12\}$. The lag, ridge, radius, posterior, loss, confidence, and chronological protocols remain unchanged. Every fixed K pays the same normalized geometric rule-count code as the radius-controlled family; selection of K is therefore not free. The one-rule case is excluded from the primary fuzzy sensitivity analysis because it is an affine degeneracy. The extended engine evaluates 16,450 eligible candidates with no build failures.

Paired series-level differences use 20,000-resample bootstrap intervals and two-sided Wilcoxon signed-rank tests with Holm correction. Rank-biserial effect sizes are reported. For energy trajectories, uncertainty is assessed with moving-block bootstrap because forecast errors are serially dependent. PJM uses a primary 14-day block with 7- and 28-day sensitivity analyses; Tetouan uses a primary seven-day block at its two-hour resolution. These analyses quantify uncertainty for the observed trajectories and do not create a new confirmatory experiment.

C. Tetouan Development Decision Case

The first operational study uses the UCI Tetouan City power-consumption data [24]. Ten-minute observations are aggregated to two-hour means for three zones and split 20/45/15/20. Candidate families are ridge, fixed eight-rule TSK,

TABLE II
SYNTHETIC DEVELOPMENT PROCESSES. EACH TRAJECTORY RETAINS 2,000 OBSERVATIONS AFTER A BURN-IN OF 500.

Process	Definition	Purpose
AR(2)	$Y_t = 0.6Y_{t-1} - 0.2Y_{t-2} + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, 0.2^2)$.	Linear short-memory dependence.
SETAR NARMA-10	$0.65Y_{t-1} + \varepsilon_t$ if $Y_{t-1} \leq 0$; otherwise $-0.45Y_{t-1} + 0.25Y_{t-2} + \varepsilon_t$. Standard tenth-order nonlinear moving-average recursion with $U_t \sim \mathcal{U}(0, 0.5)$.	Threshold nonlinearity. Nonlinear finite memory.
Mackey–Glass GARCH	Euler recursion with delay 17 and additive $\mathcal{N}(0, 0.01^2)$ noise. Zero-mean GARCH(1,1) with standardized Student- t_5 innovations.	Delayed nonlinear dynamics. Conditional heteroskedasticity and heavy tails.
Structural break	AR(1) mean, coefficient, and noise scale change after 60% of the retained sample.	Deliberate nonstationarity and clipping stress.

and radius-controlled TSK with lags $\{3, 6, 12\}$. The initial gate combines a certificate threshold, clipping control, and a validation comparison with a 12-step seasonal-naive fallback. Underforecast errors receive weight two and overforecast errors weight one. This study is developmental: its negative test outcome is retained and used only to redesign the later gate. Full gate logic and run metadata are moved to the supplementary material.

D. Internally Locked PJM Holdout Case

After the Tetouan outcome, a stricter gate was frozen before opening PJM test outcomes. Four predeclared regional hourly-load series—AEP, COMED, DAYTON, and PJME—are taken from the public Hourly Energy Consumption collection [25]. Duplicate timestamps are averaged, and the common interval from 1 January 2012 through 2 August 2018 is aggregated to 2,406 daily means per region, requiring at least 23 hourly values per retained day. No exogenous variables are used.

The split is 20/45/15/20. Candidate families are ridge, fixed eight-rule TSK, and radius-controlled TSK; lags are $\{7, 14, 28\}$ days, radii are $\{0.75, 1, 1.25, 1.5, 2, 2.5\}$, and ridge penalties are unchanged. Confidence is 0.05/4. Eligibility combines an untruncated certificate threshold of 0.10, certification clipping no larger than 2%, validation improvements of at least 5% over seven-day seasonal naive in both RMSE and asymmetric cost, and blockwise temporal-stability constraints. Among eligible candidates, validation asymmetric cost is primary. If none qualifies, the fallback is deployed.

All four decisions are serialized and hashed before test forecasts are computed. The gate, candidate family, and decisions are unchanged in all subsequent analyses. A compute-only interruption occurred before test opening and was recovered with unchanged locked inputs; the incident chronology, hashes, and restart evidence are reported only in the supplementary audit rather than in the main narrative.

E. Secondary Post-Hoc SARIMA and ETS Benchmarks

After the original PJM outcomes were opened, we added simple statistical forecasting comparisons to contextualize the seven-day seasonal-naive baseline. The analysis is explicitly secondary and post hoc: it does not modify the gate, certificates, pretest decisions, or confirmatory interpretation. For each

PJM region, a finite SARIMA grid and a finite exponential-smoothing (ETS) grid are selected using the original validation segment and asymmetric-cost criterion. The selected specification is then fitted on all pretest observations and evaluated by rolling one-step forecasts with fixed parameters and observed-target state updates. Test outcomes are not used for hyperparameter selection. Full candidate grids and per-region results are given in the supplementary material and reproducibility artifact.

F. Post-Hoc Fuzzy-Rule Inspection

To show what the learned fuzzy structures actually represent, we reconstruct two selected multi-rule development models without changing their fits: a five-rule radius TSK for SETAR seed 3001 and the four-rule Tetouan zone-1 model. Antecedent centers and spreads are transformed back to original data units, posterior-center local consequents are expressed as affine functions of lagged observations, and mean firing weights are computed on bound plus validation observations. These are interpretability summaries of deterministic point models, not additional model selection and not direct PAC-Bayesian guarantees.

G. Metrics and Interpretation

Synthetic prediction is summarized by standardized RMSE and MAE. Energy studies report raw-scale RMSE, MAE, and asymmetric absolute cost. Certificate values, empirical Gibbs upper terms, structural cost, Gaussian KL, randomized dimension, rule count, and target/forecast clipping are recorded separately. The deterministic posterior-center predictor in (7) produces the reported validation and test forecasts; the certificate directly concerns the Gibbs predictor and clipped BMA defined in Section III. A certificate is therefore treated as a screening quantity and never as direct evidence that the point predictor beats a forecasting baseline.

VI. RESULTS

A. Structural Sparsity Reduces Charged Complexity

Table III summarizes the 30-series development ablation under validation-RMSE selection. Fixed- K TSK has the largest randomized dimension, KL, and certificate. Radius control

TABLE III
DEVELOPMENT ABLATION OVER 30 SYNTHETIC SERIES. MODELS ARE SELECTED BY VALIDATION RMSE; LOWER VALUES ARE BETTER.

Family	$\bar{K}$	$\bar{D}$	Gaussian KL	Certificate
Ridge	1.00	13.4	5.16	0.1145
Fixed- K TSK	12.00	135.6	39.55	0.2221
Radius TSK	3.80	39.7	16.63	0.1589
Radius Top-3 TSK	3.57	40.2	16.83	0.1589

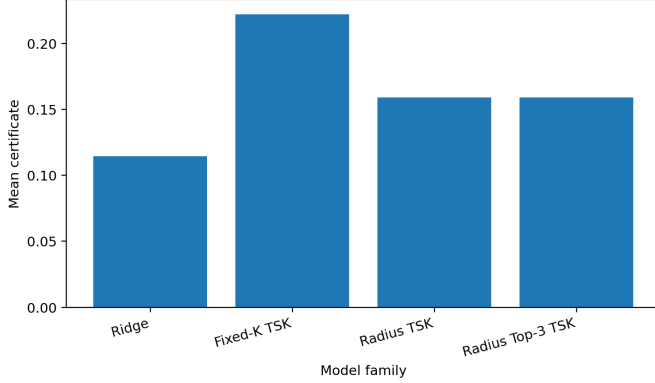


Fig. 1. Mean familywise certificate under validation-RMSE selection across 30 synthetic development series. Ridge is tightest; radius control substantially improves on the originally prespecified fixed $K = 12$ TSK baseline.

reduces the mean rule count from 12 to 3.80 and the mean dimension from 135.6 to 39.7.

The paired inference in Table IV establishes the contrast with the originally prespecified dense $K = 12$ baseline. Relative to that baseline, radius-controlled all-active TSK reduces dimension by 95.9 coefficients (95% CI $[-120.9, -71.9]$), Gaussian KL by 22.917 ($[-40.698, -10.461]$), and certificate value by 0.0631 ($[-0.0833, -0.0460]$) on average. The corresponding Holm-adjusted Wilcoxon tests remain significant, with large negative rank-biserial effects. Test RMSE is lower by 0.0335 on average ($[-0.0612, -0.0129]$). This comparison shows the cost of an unnecessarily dense rule base, but it does not by itself establish that radius selection is superior to a well-chosen smaller fixed K .

By contrast, Top-3 activation does not materially change the certificate. At the series level, the mean Top-3 minus all-active difference is -1.5×10^{-5} with 95% CI $[-0.0018, 0.0014]$ and Holm-adjusted $p = 1.000$; the test-RMSE interval also includes zero. Across 5,600 matched candidate pairs, the descriptive mean certificate difference is approximately -5×10^{-5} . Since Top-3 does not reduce $K(p+1)$, it should be described as a computational approximation, not the source of the PAC-Bayesian complexity gain.

B. Fixed- K Sensitivity Qualifies the Radius Advantage

Table V shows that the mean fixed- K certificate increases steadily with rule count, from 0.1326 at $K = 2$ to 0.2221 at $K = 12$. The radius-controlled mean is 0.1589. Consequently, radius control is not uniformly tighter than fixed rule bases: its mean certificate is looser than fixed $K = 2$ and $K = 3$,

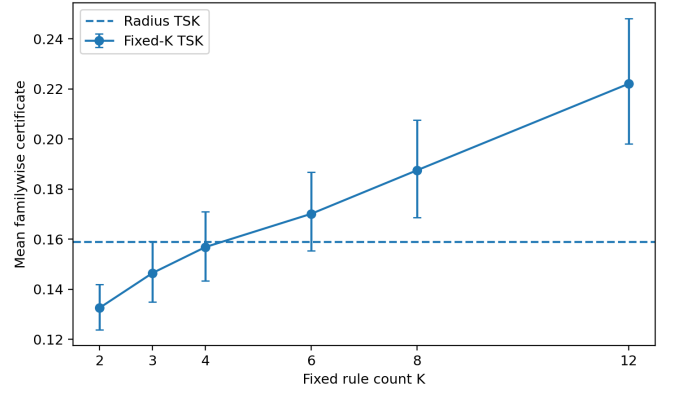


Fig. 2. Mean familywise certificate for fixed- K TSK under validation-RMSE selection. Error bars are bootstrap intervals for the per-series means; the dashed line is the radius-controlled TSK mean.

essentially matched by $K = 4$, and tighter than the larger $K = 6, 8, 12$ alternatives. After Holm correction, the paired certificate advantage is clear for $K = 8$ and $K = 12$, while the differences for $K = 3, 4, 6$ are not statistically decisive.

When each series is allowed to choose its best fixed K by validation RMSE, radius control changes the certificate by only -0.0068 on average (radius minus best fixed K ; 95% bootstrap interval $[-0.0190, 0.0047]$; Holm-adjusted $p = 0.382$). The corresponding test-RMSE mean difference is -0.0054 with bootstrap interval $[-0.0116, -0.0003]$, whereas the corrected rank test remains non-significant ($p = 0.285$). The differing conclusions reflect that the bootstrap targets the mean while the signed-rank test targets the paired rank distribution and is affected by ties and heterogeneous effects. Nine of 30 series select $K = 2$, five select $K = 3$, four select $K = 4$, six select $K = 6$, five select $K = 8$, and only one selects $K = 12$. Under certificate selection, all 30 series select fixed $K = 2$.

These results materially narrow the structural claim. Radius construction is useful as an adaptive mechanism that avoids large unnecessary rule bases, but the certificate gain relative to $K = 12$ is mainly a rule-count effect. The current evidence does not show that radius control dominates a predeclared and fairly charged small fixed- K grid.

C. The Radius–Rule–Dimension–KL Chain

Within each process, family, and lag, Spearman correlations across radius values average -0.847 for radius versus rule count, 1.000 for rule count versus dimension, 0.981 for dimension versus Gaussian KL, and 0.982 for Gaussian KL versus certificate. These are descriptive associations rather than a universal monotonicity theorem, but they make the realized complexity path observable:

$$r \longrightarrow K(r) \longrightarrow D(r) \longrightarrow \text{KL} \longrightarrow \text{Cert}. \quad (28)$$

Ridge retains the smallest mean certificate under both selection strategies: 0.1145 under validation-RMSE selection and 0.1017 under certificate selection. Under validation-RMSE selection, Radius TSK exceeds ridge by 0.0445 in certificate value (95% CI $[0.0296, 0.0609]$; Holm-adjusted $p < 10^{-8}$),

TABLE IV

PAIRED RADIUS-TSK MINUS FIXED- $K = 12$ DIFFERENCES ACROSS 30 SYNTHETIC DEVELOPMENT SERIES UNDER VALIDATION-RMSE SELECTION. CONFIDENCE INTERVALS ARE 95% PAIRED PERCENTILE-BOOTSTRAP INTERVALS FOR THE MEAN DIFFERENCE. p_H IS THE TWO-SIDED WILCOXON SIGNED-RANK p -VALUE AFTER HOLM CORRECTION ACROSS THE FOUR REPORTED ENDPOINTS; r_{rb} IS MATCHED-PAIRS RANK-BISERIAL CORRELATION. NEGATIVE VALUES FAVOR RADIUS TSK.

Metric	Mean Δ	95% CI	Radius lower	p_H	r_{rb}
Dimension	-95.9	[-120.9, -71.9]	29/30	5.4e-06	-1.000
Gaussian KL	-22.917	[-40.698, -10.461]	27/30	2.4e-05	-0.837
Certificate	-0.0631	[-0.0833, -0.0460]	28/30	4.1e-07	-0.953
Test RMSE	-0.0335	[-0.0612, -0.0129]	24/30	1.2e-04	-0.755

TABLE V

FIXED- K SENSITIVITY UNDER VALIDATION-RMSE SELECTION ACROSS 30 SYNTHETIC DEVELOPMENT SERIES. THE PAIRED DIFFERENCE IS RADIUS TSK MINUS FIXED- K TSK; NEGATIVE VALUES FAVOR RADIUS CONTROL.

K	Dimension	Gaussian KL	Certificate	Test RMSE	Δ Cert.
2	23.7	10.180	0.1326	0.8850	+0.0264
3	34.3	13.469	0.1465	0.8846	+0.0125
4	41.9	17.294	0.1568	0.8869	+0.0021
6	60.0	21.011	0.1701	0.8950	-0.0112
8	82.9	26.413	0.1875	0.8951	-0.0286
12	135.6	39.547	0.2221	0.9149	-0.0631

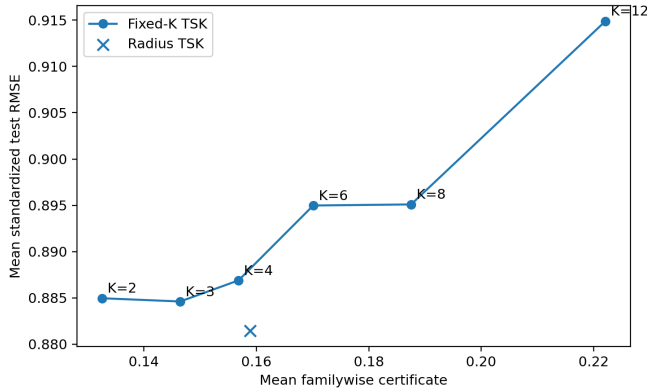


Fig. 3. Development mean test RMSE versus mean familywise certificate across fixed rule counts and radius-controlled TSK. The plot is descriptive and does not imply a universal Pareto ordering.

while its mean test-RMSE difference is small and uncertain (-0.0052 , 95% CI $[-0.0194, 0.0137]$). The results therefore do not support a claim that TSK is intrinsically easier to certify than a linear model.

D. Predictive Accuracy and Clipping

Table VI separates predictive accuracy from certificate tightness. Radius TSK improves mean test RMSE over ridge on SETAR, NARMA-10, Mackey–Glass, and the structural-break process, but is slightly worse on AR(2) and materially worse on GARCH. Its largest benefit is Mackey–Glass, where mean RMSE falls from 0.0656 to 0.0539, while its certificate is more than twice the ridge value because of additional complexity.

The structural-break process is a failure-boundary example. Certification target clipping is 23.87% and test target clipping is 71.55%. The bounded certificate remains formally meaningful

for the clipped loss, but it is not an operational description of raw errors after the break.

The localized prior improves the selected familywise certificate over a zero-mean prior in 85.6% of selected rows, with mean absolute improvement 0.0138 after charging the same prior-variant mixture cost. Localization is useful but not uniformly superior; GARCH contains counterexamples.

E. Actual Multi-Rule Fuzzy Structures

Table VII reports two reconstructed posterior-center TSK models in original units. The five SETAR rules partition different lag configurations and attach distinct local affine dynamics; for example, the dominant rule places most weight on y_{t-1} , while another assigns a negative coefficient to y_{t-1} and a larger positive coefficient to y_{t-2} . The four Tetouan rules have similar average load levels but different recent-versus-oldest prototypes and different combinations of the most recent, six-step, and twelve-step lags. This demonstrates that the implementation contains genuine local rule variation rather than repeated copies of one global affine model.

The interpretation must remain limited. These prototypes are data-driven summaries, not linguistic rules validated by domain experts. The Tetouan model also failed operationally on the held-out horizon, so interpretability does not imply predictive reliability. Full centers, spreads, local intercepts, all lag coefficients, supports, and firing frequencies are provided in the supplementary CSV artifact.

F. Negative Tetouan Development Outcome

The initial Tetouan gate admits a radius TSK model in zone 1 and a fixed- K TSK model in zone 2, while zone 3 abstains. Both learned deployments fail on the test horizon: RMSE changes are -11.86% and -1.51% , and asymmetric-cost changes are -22.67% and -19.05% . With seven-day moving blocks, the zone-1 RMSE interval is $[-24.45, -1.04]\%$ and its cost interval is $[-34.95, -10.98]\%$; zone 2 has RMSE interval $[-13.04, 8.55]\%$ and cost interval $[-35.35, -3.70]\%$. Zone 3 is identically the fallback. Averaged over all three zones, RMSE worsens by 4.46% and cost worsens by 13.91%.

This negative result demonstrates that a non-vacuous certificate plus overall validation performance is not a sufficient deployment rule under temporal shift. It motivated the stricter margin and block-stability requirements in the independent confirmatory protocol.

TABLE VI
SYNTHETIC DEVELOPMENT RESULTS SELECTED BY VALIDATION RMSE (FIVE SEEDS PER PROCESS). RMSE IS ON THE PRIOR-STANDARDIZED SCALE.

Process	Test RMSE		Certificate		Target clipping	
	Ridge	Radius TSK	Ridge	Radius TSK	Cert.	Test
AR(2)	0.9098	0.9126	0.1128	0.1271	0.62%	0.35%
SETAR	1.0140	0.9798	0.1131	0.1467	0.42%	0.50%
NARMA-10	0.7670	0.7656	0.1093	0.1188	0.97%	0.30%
Mackey–Glass	0.0656	0.0539	0.1129	0.2403	0.92%	1.30%
GARCH	1.0398	1.0926	0.0941	0.1114	0.25%	0.20%
Structural break	1.5236	1.4841	0.1445	0.2093	23.87%	71.55%

TABLE VII
POST-HOC EXAMPLES OF ACTUAL MULTI-RULE POSTERIOR-CENTER TSK MODELS. ANTECEDENT PROTOTYPES AND CONSEQUENT COEFFICIENTS ARE REPORTED IN ORIGINAL DATA UNITS. MEAN FIRING IS COMPUTED ON BOUND PLUS VALIDATION OBSERVATIONS. THESE DETERMINISTIC RULES ARE THE POINT PREDICTORS USED FOR VALIDATION/TEST EVALUATION; THE PAC-BAYES GUARANTEE DIRECTLY COVERS THE GIBBS PREDICTOR AND, BY CONVEXITY, THE CLIPPED BAYESIAN MODEL AVERAGE, NOT THESE POINT FORECASTS.

Example	Rule	Prior support	Mean firing	Center mean	Recent-oldest	Dominant local consequent terms
SETAR seed 3001	R1	143	34.1%	-0.11	0.04	+0.458 y(t-1); +0.179 y(t-2); +0.119 y(t-3)
SETAR seed 3001	R4	74	19.0%	0.06	0.24	+0.487 y(t-2); -0.422 y(t-1); -0.037 y(t-3)
SETAR seed 3001	R3	82	18.5%	-0.37	0.09	+0.685 y(t-1); +0.067 y(t-2); -0.006 y(t-3)
SETAR seed 3001	R5	61	15.3%	-0.01	-0.23	+0.183 y(t-1); +0.106 y(t-3); +0.003 y(t-2)
SETAR seed 3001	R2	37	13.1%	-0.19	-0.56	+0.576 y(t-1); -0.160 y(t-3); +0.067 y(t-2)
Tetouan Zone 1	R1	216	25.5%	31094.05	5904.95	+0.641 y(t-12); +0.463 y(t-1); +0.259 y(t-6)
Tetouan Zone 1	R2	216	25.0%	31033.57	-2317.05	+0.927 y(t-1); -0.198 y(t-2); +0.164 y(t-7)
Tetouan Zone 1	R4	216	24.9%	31018.71	-1208.12	+0.888 y(t-1); +0.459 y(t-12); -0.299 y(t-4)
Tetouan Zone 1	R3	214	24.5%	30957.31	-3487.66	+0.542 y(t-12); -0.466 y(t-10); +0.356 y(t-2)

TABLE VIII
OPERATIONAL DECISION STUDIES. IMPROVEMENTS ARE RELATIVE TO THE PREDECLARED SEASONAL-NAIVE FALLBACK; NEGATIVE VALUES INDICATE DEGRADATION.

Phase	Series	Deployment	Certificate	Test RMSE imp.	Test cost imp.
Tetouan dev.	zone 1	dense_tsk	0.1226	-11.86%	-22.67%
Tetouan dev.	zone 2	fixed_k_dense_tsk	0.1482	-1.51%	-19.05%
Tetouan dev.	zone 3	seasonal_naive_24h	—	0.00%	0.00%
PJM confirm.	AEP	ridge	0.0959	57.97%	58.64%
PJM confirm.	COMED	ridge	0.0841	55.40%	54.11%
PJM confirm.	DAYTON	dense_tsk	0.0933	49.13%	50.30%
PJM confirm.	PJME	ridge	0.0959	51.70%	51.42%

TABLE IX
SECONDARY PAIRED MOVING-BLOCK-BOOTSTRAP UNCERTAINTY INTERVALS FOR THE FROZEN PJM TEST IMPROVEMENTS. CIRCULAR BLOCKS OF 14 DAYS AND 20,000 RESAMPLES ARE USED; 7- AND 28-DAY SENSITIVITY RESULTS ARE PROVIDED IN THE ARTIFACT. THE ANALYSIS WAS ADDED AFTER THE FROZEN TEST OUTCOMES AND DOES NOT ALTER THE DECISIONS.

Region	RMSE improvement [95% CI]	Cost improvement [95% CI]
AEP	57.97% [53.06, 61.61]	58.64% [54.00, 62.43]
COMED	55.40% [49.58, 59.83]	54.11% [48.31, 58.81]
DAYTON	49.13% [43.81, 53.53]	50.30% [45.16, 54.74]
PJME	51.70% [45.80, 56.29]	51.42% [45.71, 56.34]

G. Independent PJM Confirmatory Outcome

The frozen PJM gate admits a learned model in all four regions. Test RMSE improves by 57.97%, 55.40%, 49.13%, and 51.70% for AEP, COMED, DAYTON, and PJME, respectively. Corresponding asymmetric-cost improvements are 58.64%, 54.11%, 50.30%, and 51.42%; mean improvements are 53.55%

and 53.62%. Table IX reports secondary 14-day moving-block-bootstrap intervals. Every lower bound remains positive, and the same conclusion holds under 7- and 28-day block-length sensitivity analyses. Selected certificates range from 0.0841 to 0.0959, and test target and forecast clipping are zero.

The confirmatory result does not establish nonlinear fuzzy superiority. AEP, COMED, and PJME select ridge. DAYTON selects a radius-controlled one-rule TSK with lag 14; with one rule, this is an affine predictor. No fixed-eight-rule TSK candidate passes the frozen gate. The operational evidence therefore supports a complexity-aware gate that can shrink to a simple model, not a claim that a multi-rule TSK system caused the gain.

H. Secondary Post-Hoc Statistical Benchmarks

The post-hoc SARIMA and ETS comparisons provide a stricter context for the PJM result. Across the four regions, selected SARIMA specifications improve RMSE over the seasonal-naive fallback by 56.82% on average and improve

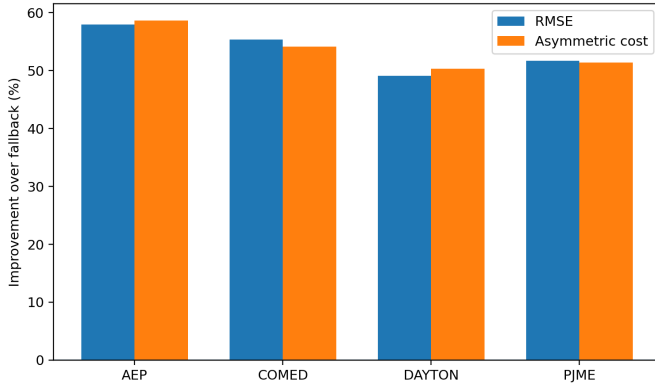


Fig. 4. Independent PJM test improvements over the frozen seven-day seasonal-naive fallback.

on the originally deployed point forecasts by 6.99%. ETS improves over the fallback by 54.77% on average and over the deployed forecasts by 2.43%. SARIMA has lower test RMSE than the deployed model in all four regions; ETS is better in AEP and DAYTON, essentially tied in COMED, and worse in PJME. The full per-region table is reported in the supplementary material.

These comparisons were designed after the original PJM outcomes were opened. They therefore cannot be used to revise the frozen gate or strengthen the original internally locked claim. Instead, they show that the large gain over seasonal naive is not unique to the PAC-Bayesian candidate family and that simple statistical forecasting models remain competitive or superior. The operational contribution is consequently the auditable screening and abstention protocol, not state-of-the-art point forecasting.

VII. DISCUSSION AND LIMITATIONS

A. An Integration Result, Not a Model-Superiority Result

The strongest contribution of the study is methodological integration. The framework places chronological prior construction, complete structural coding, Gaussian posterior localization, bounded sequential risk, model selection, and operational abstention inside one auditable pipeline. Each element is individually established or technically standard; the contribution lies in making their interaction explicit and reproducible for linear and TSK forecasters under temporal dependence.

The structural experiments support a precise but limited mechanism. Replacing an unnecessarily dense 12-rule TSK model with a smaller rule base reduces randomized dimension, Gaussian KL, and certificate value. The Fixed- K sensitivity analysis shows, however, that small predeclared fixed rule bases can match or improve on radius control. The supported conclusion is therefore that removal of complete rule blocks matters for the charged complexity. The present evidence does not establish that radius construction is intrinsically superior to every alternative way of choosing a compact rule base.

Top-3 activation provides a complementary negative result. It changes the number of rules evaluated at a sample but leaves the randomized consequent dimension unchanged. Its negligible

certificate effect is therefore consistent with the mathematical accounting. For certificate-aware fuzzy design, sparsity must operate on the parameterized rule structure or on a posterior over rule inclusion, not only on firing-strength truncation at prediction time.

B. Interpretability and Its Boundary

The reconstructed SETAR and Tetouan rules show that the implementation learns distinct local affine relationships associated with different antecedent prototypes. This is useful for auditing the model and checking that a multi-rule TSK system is not merely duplicating one global regression. It is not evidence that the prototypes correspond to physically validated regimes, and it is not evidence that interpretability guarantees robustness. The Tetouan multi-rule model is the clearest warning: its rules are inspectable, yet the deployment decision fails on the held-out horizon.

The operational point forecasts are deterministic posterior-center predictions, whereas the PAC-Bayesian guarantee directly covers the Gibbs predictor and, by convexity, the clipped Bayesian model average. Before clipping, the posterior mean of a linear-in-consequents TSK predictor equals the posterior-center prediction. After clipping, equality with the clipped BMA generally fails. The certificate is therefore used as a screening statistic rather than represented as a direct guarantee for the deployed point forecast.

C. Operational Evidence and Stronger Baselines

The negative Tetouan study and positive internally locked PJM holdout illustrate why a certificate alone is not a deployment rule. The redesigned PJM gate combines certificate thresholding with validation margins and blockwise temporal-stability checks. Moving-block-bootstrap intervals remain positive for the original PJM comparison with seasonal naive, but these intervals quantify uncertainty only for the observed trajectories and do not prove population-level transfer to other regions or future regimes.

The secondary SARIMA and ETS analysis materially narrows the operational interpretation. SARIMA improves on the originally deployed point forecast in all four PJM regions, while ETS is competitive and sometimes better. Because these comparisons were designed after the PJM outcomes were opened, they are post hoc and cannot replace the frozen protocol. They nevertheless show that the gain over seasonal naive is not specific to TSK or PAC-Bayesian selection. The defensible operational contribution is an auditable process for screening, margin checking, and abstention, not superior forecasting accuracy relative to established statistical methods.

D. Limits of the Certificate

The certificate controls average conditional risk of a clipped Gibbs predictor over the certification horizon and transfers to the clipped BMA. It does not control raw RMSE, the deterministic posterior-center predictor, an unseen future horizon, or relative performance against another forecasting method. Structural-break experiments demonstrate the practical cost

of the bounded-loss construction: when a prior-fixed clipping interval is exceeded frequently, the certificate remains formally valid for the clipped loss but loses relevance to raw-scale errors.

Heavy-tail or variance-adaptive supermartingale bounds [14], [15] are a principled next step, but they require a separately verified implementation. The time-uniform proposition is evaluated numerically only at the final certification horizon; the paper does not claim an executed confidence sequence over all prefixes.

E. Remaining Empirical and Procedural Limitations

The synthetic study uses five seeds per process. The paired analysis is adequate for large structural contrasts such as Radius TSK versus fixed $K = 12$, but it is not sufficient to characterize all nonlinear regimes or small performance differences. The PJM case contains four regional series over one common historical interval and uses no exogenous variables. The post-hoc statistical grid is intentionally simple and does not cover gradient boosting, modern probabilistic forecasting, or deep sequence models.

The PJM protocol is internally locked and supported by hashes, manifests, and pretest decision files, but it was not externally preregistered. A compute-only interruption occurred before test opening; recovery used unchanged locked inputs. The complete chronology is retained in the supplementary audit. These controls improve traceability but are not equivalent to an external timestamped preregistration.

F. Implications for Future Fuzzy-System Research

A stronger fuzzy contribution would replace deterministic radius construction with a posterior over rule inclusion or a certificate-aware structural learning objective. Such a method could jointly trade predictive fit, interpretable rule blocks, and charged complexity. It should be compared fairly with small fixed- K grids and with strong statistical and machine-learning baselines. Domain-expert validation of learned rule semantics would also be needed before presenting local prototypes as operational explanations rather than model diagnostics.

VIII. CONCLUSION

This work presents an auditable framework that combines chronological prior construction, explicit structural coding, a computable sequential PAC-Bayesian certificate, linear and TSK forecasting families, and conservative deployment screening. The general supermartingale inequality is established in prior work; the contribution is its fully charged, reproducible specialization and the empirical separation of different kinds of sparsity.

The structural experiments show that removing complete TSK rule blocks can reduce randomized dimension, Gaussian KL, and certificate value relative to an unnecessarily dense rule base. The additional Fixed- K sensitivity prevents a stronger claim: small fixed rule bases can match or tighten the radius-controlled certificate, so the evidence supports structural compactness rather than universal superiority of radius selection. Sparse activation alone does not deliver the same complexity

reduction when all consequent coordinates remain randomized and charged.

The energy studies show that a valid certificate can contribute to a broader screening rule but cannot replace forecasting validation. The first Tetouan gate fails. The redesigned PJM gate succeeds relative to its predeclared seasonal-naïve fallback, yet the selected models are linear or one-rule affine predictors. Secondary post-hoc SARIMA and ETS benchmarks often improve further on the deployed point forecasts. Accordingly, the real-data evidence supports transparent model screening, validation margins, temporal-stability checks, and abstention; it does not establish nonlinear fuzzy-model dominance or state-of-the-art forecasting accuracy.

The main conclusion is therefore modest: PAC-Bayesian complexity can be made explicit and auditable for linear and TSK time-series models, and true structural rule reduction can tighten that accounting. Future work should learn a posterior over rule inclusion, align the certified and operational prediction objects more directly, compare against stronger forecasting baselines in a new externally timestamped protocol, and replace clipping with verified heavy-tail or variance-adaptive sequential guarantees.

DATA AND CODE AVAILABILITY

The source code, processed data, experimental configurations, result tables, audit manifests, and supplementary materials supporting this study are publicly available at <https://github.com/muneeff/pacbayes-sparse-tsk>.

The exact version associated with this manuscript is archived as release v4.3.1: <https://github.com/muneeff/pacbayes-sparse-tsk/releases/tag/v4.3.1>.

Raw third-party datasets are not redistributed. Download, verification, and preprocessing scripts are provided in the repository.

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The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AND CODE AVAILABILITY

The complete code, frozen configurations, raw result tables, pre-outcome locks, and audit manifests are included in the accompanying reproducibility artifact. The public repository URL will be inserted in the final submission; the review package uses an anonymized archive.

REFERENCES

- [1] B. Chugg, H. Wang, and A. Ramdas, “A unified recipe for deriving (time-uniform) PAC-Bayes bounds,” *Journal of Machine Learning Research*, vol. 24, 2023.
- [2] L. Ralaivola, M. Szafranski, and G. Stempfel, “Chromatic PAC-Bayes bounds for non-IID data: Applications to ranking and stationary beta-mixing processes,” *Journal of Machine Learning Research*, vol. 11, 2010.
- [3] P. Alquier and O. Wintenberger, “Model selection for weakly dependent time series forecasting,” *Bernoulli*, vol. 18, 2012.
- [4] P. Alquier, X. Li, and O. Wintenberger, “Prediction of time series by statistical learning: General losses and fast rates,” *Dependence Modeling*, vol. 1, 2013.
- [5] I. Banerjee, V. A. Rao, and H. Honnappa, “PAC-Bayes bounds on variational tempered posteriors for Markov models,” *Entropy*, vol. 23, no. 3, p. 313, 2021.
- [6] D. Eringis, J. Leth, Z.-H. Tan, R. Wisniewski, and M. Petreczky, “PAC-Bayes generalisation bounds for dynamical systems including stable RNNs,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024.
- [7] Y. Seldin, F. Laviolette, N. Cesa-Bianchi, J. Shawe-Taylor, and P. Auer, “PAC-Bayesian inequalities for martingales,” *IEEE Transactions on Information Theory*, vol. 58, 2012.
- [8] M. Haddouche and B. Guedj, “Online PAC-Bayes learning,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022.
- [9] E. Parrado-Hernandez, A. Ambroladze, J. Shawe-Taylor, and S. Sun, “PAC-Bayes bounds with data dependent priors,” *Journal of Machine Learning Research*, vol. 13, 2012.
- [10] O. Rivasplata, I. Kuzborskij, C. Szepesvari, and J. Shawe-Taylor, “PAC-Bayes analysis beyond the usual bounds,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 16 833–16 845.
- [11] M. Perez-Ortiz, O. Rivasplata, J. Shawe-Taylor, and C. Szepesvari, “Tighter risk certificates for neural networks,” *Journal of Machine Learning Research*, vol. 22, 2021.
- [12] A. Y. K. Foong, W. P. Bruinsma, D. R. Burt, and R. E. Turner, “How tight can PAC-Bayes be in the small data regime?” in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 4093–4105.
- [13] M. Haddouche, B. Guedj, O. Rivasplata, and J. Shawe-Taylor, “PAC-Bayes unleashed: Generalisation bounds with unbounded losses,” *Entropy*, vol. 23, no. 10, p. 1330, 2021.
- [14] M. Haddouche and B. Guedj, “PAC-Bayes generalisation bounds for heavy-tailed losses through supermartingales,” *Transactions on Machine Learning Research*, 2023.
- [15] B. Rodriguez-Galvez, R. Thobaben, and M. Skoglund, “More PAC-Bayes bounds: From bounded losses, to losses with general tail behaviors, to anytime validity,” *Journal of Machine Learning Research*, vol. 25, 2024.
- [16] X. Gu and S. Wang, “Bayesian Takagi-Sugeno-Kang fuzzy model and its joint learning of structure identification and parameter estimation,” *IEEE Transactions on Industrial Informatics*, vol. 14, no. 12, 2018.
- [17] J. Li, C. Hua, Y. Yang, and X. Guan, “Bayesian block structure sparse based T-S fuzzy modeling for dynamic prediction of hot metal silicon content in the blast furnace,” *IEEE Transactions on Industrial Electronics*, vol. 65, no. 6, 2018.
- [18] —, “Data-driven Bayesian-based Takagi-Sugeno fuzzy modeling for dynamic prediction of hot metal silicon content in blast furnace,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 2, 2022.
- [19] E. Lughofer and S. Kindermann, “SparseFIS: Data-driven learning of fuzzy systems with sparsity constraints,” *IEEE Transactions on Fuzzy Systems*, vol. 18, no. 2, 2010.
- [20] M. Luo, F. Sun, and H. Liu, “Hierarchical structured sparse representation for T-S fuzzy systems identification,” *IEEE Transactions on Fuzzy Systems*, vol. 21, no. 6, 2013.
- [21] K. Wiktorowicz and T. Krzeszowski, “Identification of time series models using sparse Takagi-Sugeno fuzzy systems with reduced structure,” *Neural Computing and Applications*, vol. 34, no. 10, 2022.
- [22] S. L. Chiu, “Fuzzy model identification based on cluster estimation,” *Journal of Intelligent and Fuzzy Systems*, vol. 2, no. 3, pp. 267–278, 1994.
- [23] P. P. Angelov and D. P. Filev, “An approach to online identification of Takagi-Sugeno fuzzy models,” *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 34, no. 1, 2004.
- [24] UCI Machine Learning Repository, “Power Consumption of Tetouan City,” 2017. [Online]. Available: <https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city>
- [25] R. Mulla, “Hourly Energy Consumption,” Kaggle dataset, 2018. [Online]. Available: <https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption>